

Test on Linear Algebra. May 30 2023

1. Consider a set of hermitian matrices in $M_2(\mathbb{C})$ which we denote by $\mathcal{H}_2 = \{A \in M_2(\mathbb{C}) \mid A^T = \overline{A}\}$.
 - Prove that $\mathcal{H}_2$ is a subspace in the real vector space $M_2(\mathbb{C})$ of dimension 8.
 - Prove that $\mathcal{H}_2$ is not a subspace in the complex vector space $M_2(\mathbb{C})$ of dimension 4.
 - Find a dimension of the real subspace $\mathcal{H}_2 \subset M_2(\mathbb{C})$.
2. Consider a subspace $U = \text{Span} \left(\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right)$ in the euclidean space $\mathbb{R}^4$ equipped with a standard dot product. Find the dimension of orthogonal complement $U^\perp$.
3. Consider two linear operators on the Euclidean space $\mathbb{R}^3$ equipped with the standard dot product. One is given by $L_A(\mathbf{x}) = A\mathbf{x}$ and another is given by $L_B(\mathbf{x}) = \mathbf{x}$ where $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$.
 - Prove that both operators are isometries.
 - Detect which of L_A and L_B is a rotation and which is reflection in the plane?
 - What is the axis and angle of the rotation? Write the direction vector of the axis.
 - What is a mirror of the reflection? Write the equation of the plane.